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1. Problem Statement

Modern students and researchers increasingly rely on Generative AI and Large Language Models (LLMs) to synthesize dense academic course materials, problem sets, and textbooks. However, general-purpose LLM chat interfaces present three critical limitations in academic settings:

- **Hallucination & Lack of Verifiability:** AI models routinely produce plausible-sounding yet factually inaccurate statements without providing auditable, page-level citations or visual document evidence.
- **The Speculation Trap (No Refusal Cutoff):** When course materials do not contain the answer, conventional chatbots speculate using general training data rather than admitting the answer is absent from the syllabus.
- **Context Loss & Prompt Injection Vulnerability:** General interfaces fail to isolate user-uploaded documents from core instructions, risking syllabus confusion or adversarial prompt injection.

2. Why This Problem Statement Was Chosen

Academic study demands strict intellectual accountability: *"Don't just answer the student; show them exactly where the answer came from."*

"When university students prepare for exams or technical coursework, ungrounded speculation is worse than no answer. A student needs deterministic proof, exact equation citations, and the ability to verify original pages with confidence."

- **Pedagogical Rigor:** In STEM courses (physics, linear algebra, engineering), answers must cite exact formula derivations and test parameters rather than hand-wavy approximations.
- **Verifiable Visual Proof:** Combining conversational intelligence with high-resolution PDF page rasterization allows immediate visual confirmation of underlying diagrams, charts, and equations.
- **Editorial Focus:** Designed with an academic, print-monograph reading aesthetic that promotes focused, distraction-free scholarship.



Figure 1: Interactive paper-crafted landing cover with dynamic 3D physics and typographic branding.

3. Approach & Proposed Solution

Vellum implements an end-to-end, production-grade **Retrieval-Augmented Generation (RAG)** architecture engineered specifically for provenance, explainability, and mathematical problem solving.

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[Course PDFs / Lecture Notes] ? ? core/document_processor.py ??? Page-aware extraction, OCR checks, hyphenation
cleanup ? ? core/chunker.py ??? Paragraph-aware chunking with overlap & page provenance ? ? core/embeddings.py ???
Google gemini-embedding-001 (RETRIEVAL_DOCUMENT / QUERY) ? ? core/vector_store.py ??? ChromaDB persistent vector
database (SHA-256 deduplication) ? ? (Vector Similarity Search) ? core/retriever.py ??? Top-K candidate extraction,
hybrid lexical boost & threshold ? ? core/llm.py (Rewriter) ??? Contextual query rewriting (gemini-3.5-flash-lite)
? ? [Student Question]
```

- **Paragraph-Aware Chunking:** Documents are split along natural paragraph boundaries (`\n\n`), preserving intact formulas and page metadata (`page_number`, `chunk_id`, `document_id`).
- **Smart Hybrid Retrieval:** Combines dense cosine similarity with lexical keyword overlap. Conceptual and analytical queries (theorems, formulas, problem sets) maintain semantic alignment, while off-topic queries are strictly filtered.
- **Deterministic Refusal:** If no candidate clears the calibrated relevance threshold (`MIN_RELEVANCE = 0.65`), the LLM generation is bypassed entirely, returning an honest refusal message.
- **Step-by-Step Problem Solving:** Prompts are engineered to extract given variables from problem sets and guide students through step-by-step mathematical procedures using $\LaTeX$ typesetting.



Figure 2: Three-column academic workspace featuring Knowledge Base (left), Discussion timeline (center), and Evidence panel (right).

4. Technologies & Tools Used

Layer / Component	Technology	Role in Project
Frontend & Interface	Streamlit (v1.65+)	Three-column reactive layout, chat streaming, session state management
Typography & Design	CSS3, Cormorant Garamond, Manrope	Print-monograph parchment styling, custom MutationObserver CSS bridge
Language Model	Google Gemini (gemini-3.5-flash-lite)	Conversational query rewriting, document insights, grounded answer generation
Embedding Engine	Google gemini-embedding-001	768-dimensional text vectors with task-specific retrieval tuning
Vector Database	ChromaDB	Persistent on-disk vector store with cosine distance indexing & metadata filters
PDF Processing	PyMuPDF (pymupdf / fitz)	Page-level extraction, scanned page detection, and high-DPI page rendering
Testing & QA	PyTest (43 tests)	Comprehensive test suite covering chunking, embeddings, retrieval, and UI

5. Key Features & Functionality

Continuous 3-Column Desk Organized desktop keeping library, conversation, and evidence citations simultaneously accessible without context switching.	Automated Document Insights Generates high-yield summaries and 3 clickable starter study questions tailored to uploaded problem sets or lecture slides.
Context-Aware Query Rewriting Follow-ups (e.g. <i>"Explain that step in simpler terms"</i>) are automatically rewritten into self-contained standalone search queries.	Visual Proof & Page Render The Evidence panel renders the exact rasterized PDF page image, letting students verify original layout and equations visually.
Step-by-Step Solutions Formulates math problems (matrices, eigenvalues, wave equations) with complete step-by-step $\LaTeX$ solutions and citations [1], [2].	Notes Export & Cloud Ready One-click export of chat sessions into formatted Markdown study notes. Built-in support for Streamlit Cloud secrets management.